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## Education

**PhD, Mathematics**, Dartmouth College, Hanover, NH Fall 2018 - Spring 2023

Advisor: John Voight

Thesis: *Counting elliptic curves with a cyclic  $m$ -isogeny over  $\mathbb{Q}$*

**MA, Mathematics**, Dartmouth College, Hanover, NH Fall 2018 - Winter 2020

Advisor: John Voight

**MS, Mathematics**, Brigham Young University, Provo, UT Winter 2017 - Summer 2018

Advisors: Michael Griffin and Paul Jenkins

Thesis: *The arithmetic of modular grids*

**BS, Mathematics**, Brigham Young University, Provo, UT Fall 2013 - Fall 2016

## Professional Experience

**AI/ML Engineering Manager at ARKA** Winter 2025 - Present

*My responsibilities are threefold: I innovate in AI and algorithms as an engineer, serve as line manager for both junior and senior AI/ML developers, and act as my company's Algorithm SME. As an engineer, I do cutting edge work on LLMs, with networks, in ELINT, and more. I also cowrite program proposals, serve as a tech lead for a small team, and architect best practices in AI across my program. As a line manager, I nurture a personal rapport with my people, advocate for their interests, and facilitate their work on projects that engage them and enable them to grow in the ways they wish. I am also on the front lines of company recruitment. As Algorithm SME, I liaise with project leads to provide technical support and connect them with the best employees for their problems, organize and deliver presentations on both foundational and advanced algorithm topics, and serve as our company's algorithms SME.*

**Principal AI/ML Engineer at ARKA** Summer 2023 - Winter 2025

*I spearheaded the development of advanced AI solutions across multiple projects, innovating in areas such as cognitive agents, signal segmentation systems, and satellite scheduling algorithms. I conducted novel research within an Agile framework and served as Scrum Master and AI/Algorithms Lead on one project. I delivered dozens of presentations to company leadership and peers on myriad computational and mathematical topics, from automata to quantum computing.*

**CCR-Princeton SCAMP Summer Associate** Summer 2022

*I produced research in the discipline of advanced cryptography. My work included design and vulnerability analysis of novel cryptographic protocols, as well as practical and theoretical work on implementations of homomorphic encryption and secure multi-party computation.*

**NSA Graduate Mathematics Program Intern** Summer 2021

*I produced research focused on distinguishing between human and synthetic audio, using techniques from machine learning, probability theory, and topological data analysis.*

**Academic Researcher at BYU and Dartmouth** Fall 2013 - Spring 2023

*At BYU, I proved results about modular forms and algorithms on graphs. At Dartmouth, I instigated successful research into summing divergent series, the Prisoner's Dilemma, and the Ramanujan-Robin*

*criterion. My thesis develops the arithmetic statistics of elliptic curves with cyclic  $m$ -isogeny. See Research Publications and Research Preprints below.*

## **Math Instructor at BYU and Dartmouth**

Winter 2017 - Spring 2023

*I taught 30-student calculus courses in three different styles: lecturing, virtually with a “flipped classroom”, and with handouts and group exercises. I TAed for lower-division courses, and by special request for upper-division courses as well. See Teaching Experience below.*

## Computer Skills

### **Fluent in...**

Bash • Git •  $\text{\LaTeX}$  • Python • SageMath • Vim

### **Fair at...**

C++ • Go • HTML • Java • Magma • Mathematica • MatLab • R

## Research Publications

### **A family of analogues to the Robin criterion, with Steve Fan and Mits Kobayashi**

The Ramanujan Journal **70**, article 40 (2026). arXiv: 2511.02106

The Robin criterion states that the Riemann hypothesis is equivalent to the inequality  $\sigma(n) < e^\gamma n \log \log n$  for all  $n > 5040$ , where  $\sigma(n)$  is the sum of divisors of  $n$ , and  $\gamma$  is the Euler–Mascheroni constant. Define  $\sigma^{[k]}(n) := \sum_{[d_1, \dots, d_k] = n} d_1 \cdots d_k$ , where  $[d_1, \dots, d_k]$  is the least common multiple of  $d_1, \dots, d_k$ . These functions behave asymptotically like  $\sigma(n)^k$  as  $k \rightarrow \infty$ . We prove the following analogue of the Robin criterion: for any  $k \geq 2$ , the Riemann hypothesis holds if and only if  $\sigma^{[k]}(n) < \frac{(e^\gamma n \log \log n)^k}{\zeta(k)}$  for all  $n > 2162160$ , where  $\zeta$  is the Riemann zeta function.

### **Positive spoof Lehmer factorizations, with Guntas Singh**

Integers **26**, A33 (2026). arXiv: 2409.17076

We investigate the integer solutions of Diophantine equations related to Lehmer’s totient conjecture. We give an algorithm that computes all nontrivial positive spoof Lehmer factorizations with a fixed number of bases  $r$ , and enumerate all nontrivial positive spoof Lehmer factorizations with 6 or fewer factors.

### **A database of basic numerical invariants of Hilbert modular surfaces, with Eran Assaf, Angelica Babei, Ben Breen, Edgar Costa, Juanita Duque-Rosero, Aleksander Horawa, Jean Kieffer, Avinash Kulkarni, Sam Schiavone, and John Voight**

Contemp. Math., vol. 796, 2024, Amer. Math. Soc., Providence, RI, 285–312. arXiv: 2301.10302

We describe algorithms for computing geometric invariants for Hilbert modular surfaces, and we report on their implementation.

### **Counting elliptic curves over the rationals with a 7-isogeny, with John Voight**

Research in Number Theory **9**, 75 (2023). arXiv: 2212.11354

We count by height the number of elliptic curves over the rationals, both up to isomorphism over the rationals and over an algebraic closure thereof, that admit a cyclic isogeny of degree 7.

**Reactive means in the Iterated Prisoner's Dilemma**, with *Caroline Hammond and Feng Fu*  
Applied Mathematics and Computation **458** 128201 (2023). arXiv: 2302.13909

The Iterated Prisoner's Dilemma (IPD) is a well studied framework for understanding direct reciprocity and cooperation in pairwise encounters. However, measuring the morality of various IPD strategies is still largely lacking. Here, we partially address this issue by proposing a suit of plausible morality metrics to quantify four aspects of justice. We focus our closed-form calculation on the class of reactive strategies because of their mathematical tractability and expressive power. We define reactive means as a tool for studying how actors in the IPD and Iterated Snowdrift Game (ISG) behave under typical circumstances. We compute reactive means for four functions intended to capture human intuitions about "goodness" and "fair play". Two of these functions are strongly anticorrelated with success in the IPD and ISG, and the other two are weakly anticorrelated with success. Our results will aid in evaluating and comparing powerful IPD strategies based on machine learning algorithms, using simple and intuitive morality metrics.

**7 internal NSA papers**, with *IDA Research Team*

Summer 2022.

Abstracts redacted.

**9 internal IDA blog posts**

Summer 2022.

Abstracts redacted.

**Generating and detecting synthetic speech: a technical report**, with *NSA Research Team*  
Internal NSA Paper (Summer 2021).

We present methods for distinguishing between human and synthesized audio, assess their efficacy, and suggest directions for future development. We also present methods for generating realistic-sounding synthetic speech that mimics a human voice, and for confusing machine learning systems such as speaker ID and detection algorithms into believing the generated synthetic speech is real.

**The arithmetic of modular grids**, with *Michael Griffin and Paul Jenkins*

Mathematika **68**, 1080-1119 (2022). arXiv: 2012.14403

A modular grid is a pair of sequences  $(f_m)_m$  and  $(g_n)_n$  of weakly holomorphic modular forms such that for almost all  $m$  and  $n$ , the coefficient of  $q^n$  in  $f_m$  is the negative of the coefficient of  $q^m$  in  $g_n$ . Zagier proved this coefficient duality in weights  $1/2$  and  $3/2$  in the Kohnen plus space, and such grids have appeared for Poincaré series, for modular forms of integral weight, and in many other situations. We give a general proof of coefficient duality for canonical row-reduced bases of spaces of weakly holomorphic modular forms of integral or half-integral weight for every group  $\Gamma \subseteq \mathrm{SL}_2(\mathbb{R})$  commensurable with  $\mathrm{SL}_2(\mathbb{Z})$ . We construct bivariate generating functions that encode these modular forms, and study linear operations on the resulting modular grids.

**Odd, spoof perfect factorizations**, with *Nickolas Andersen, Spencer Durham, Michael Griffin, Jonathon Hales, Paul Jenkins, Ryan Keck, Hankun Ko, Eric Moss, Pace Nielsen, Kyle Niendorf, Vandy Tombs, Merrill Warnick, and Dongsheng Wu*

Journal of Number Theory **234**, 31-47 (2022). Quanta article. Veritasium video. arXiv: 2006.10697

We investigate the integer solutions of Diophantine equations related to perfect numbers. These solutions generalize the example, found by Descartes in 1638, of an odd, "spoof" perfect factorization  $3^2 \cdot 7^2 \cdot 11^2 \cdot 22021^1$ . More recently, Voight found the spoof perfect factorization  $3^4 \cdot 7^2 \cdot 11^2 \cdot 19^2 \cdot (-127)^1$ . No other examples appear in the literature. We compute all nontrivial, odd, primitive spoof perfect factorizations with fewer than seven bases – there are twenty-one in total. We show that the structure of odd, spoof perfect factorizations is extremely rich, and there are multiple infinite families of them. This implies that certain approaches to the odd perfect number problem that use only the multiplicative nature of the sum-of-divisors function are unworkable. On the other hand, we prove that there are only finitely many nontrivial, odd, primitive spoof perfect factorizations with a fixed number of bases.

**Zagier duality for level  $p$  weakly holomorphic modular forms**, with *Paul Jenkins*

The Ramanujan Journal **50**, 93–109 (2019). arXiv: 1709.10023

We prove Zagier duality between the Fourier coefficients of canonical bases for spaces of weakly holomorphic modular forms of prime level  $p$  with  $11 \leq p \leq 37$  with poles only at the cusp at  $\infty$ , and special cases of duality for an infinite class of prime levels. We derive generating functions for the bases for genus 1 levels.

## Graphs with the strong Havel-Hakimi property, with Michael Barrus

Graphs and Combinatorics **32**, 1689–1697 (2016). arXiv: 1505.00085

The Havel–Hakimi algorithm iteratively reduces the degree sequence of a graph to a list of zeroes. As shown by Favaron, Mahéo, and Saclé, the number of zeroes produced, known as the residue, is a lower bound on the independence number of the graph. We say that a graph has the strong Havel–Hakimi property if in each of its induced subgraphs, deleting any vertex of maximum degree reduces the degree sequence in the same way that the Havel–Hakimi algorithm does. We characterize graphs having this property (which include all threshold and matrogenic graphs) in terms of minimal forbidden induced subgraphs. We further show that for these graphs the residue equals the independence number, and a natural greedy algorithm always produces a maximum independent set.

## Books

### Puzzle and Proof: A Decade of Problems from the Utah Math Olympiad, with Samuel Dittmer, Hiram Golze, and Caleb Stanford

AK Peters/CRC Recreational Mathematics Series.

This book is a collection of 70 mathematical problems and puzzles from the first ten years of the Utah Math Olympiad (UMO), 2013–2022. These problems are distinguished in two respects. First, they aim to be *understandable* to an advanced high school audience, even if solving them can sometimes be quite difficult. Second, all of the problems ask for not only an answer, but a *proof*.

## Research Preprints

### The Tropical Algebra of Binary-Tree Height

Preprint. arXiv: 2608.12401

The binary-tree height recursion defines an algebra  $\mathcal{H}$  on  $\mathbb{N} \cup \{-\infty\}$ , with join given by  $\max$  and product  $a \star b = \max\{a, b\} + 1$ . We show that weighted evaluation of a labelled tree depends only on the greatest depth of each label. Single-tree profiles are exactly the vectors satisfying the binary Kraft inequality, while finite joins realize every vector in  $(\mathbb{N} \cup \{-\infty\})^n$ ; hence the  $n$ -variable term operations form the free algebra  $\mathcal{H}^n$ . We also classify  $\mathcal{H}$ 's compatible semilattice operation, subalgebras, endomorphisms, congruences, and finite quotients, and recover the dyadic-composition spectrum at the full-linear boundary.

### The Gregory function and its completed Gregory transform

Preprint. arXiv: 2608.05189

We study the entire interpolation  $\mathcal{G}(z) = \int_0^1 \binom{x}{z} dx$  of the Gregory coefficients. Its completion satisfies the positive Markov-transform identity  $\frac{\pi z}{\sin(\pi z)} \mathcal{G}(z) = \sum_{n=1}^{\infty} \frac{n|G_n|}{n-z}$ . Consequently, every zero is real and simple; the negative zeros are the integers  $-1, -2, \dots$ , and one zero  $\rho_n$  lies in each  $(n, n+1)$ . We derive complete logarithmic asymptotics for  $\rho_n - n$ , determine the Cartwright growth and canonical products of  $\mathcal{G}$ , and realize  $1/\rho_n$  spectrally. The resulting relative determinant yields  $\gamma = \sum_{n=1}^{\infty} (1/n - 1/\rho_n)$ .

### A weighted angle distance on strings

Preprint. arXiv: 2604.20633

We define a multi-scale metric  $d_\rho$  on strings by aggregating angle distances between all  $n$ -gram count vectors with exponential weights  $\rho^n$ . We benchmark  $d_\rho$  in DBSCAN clustering against edit and  $n$ -gram baselines, give a linear-time suffix-tree algorithm for evaluation, prove metric and stability properties, and characterize isometries.

### Multiplicative summations into algebraically closed fields, with Robert Dawson

Preprint. arXiv: 2111.09938

In this paper, extending our earlier program, we derive maximal canonical extensions for multiplicative summations into algebraically closed fields. We show that there is a well-defined analogue to minimal polynomials for a series algebraic over a ring of series, the “scalar polynomial”. When that ring is the domain of a summation  $\mathfrak{S}$ , we derive the related concepts of the  $\mathfrak{S}$ -minimal polynomial for a series, which is mapped by  $\mathfrak{S}$  to a scalar polynomial. When the scalar polynomial for a series has the form  $(t - a)^n$ ,  $a$  is the unique value to which the series can be mapped by an extension of the original summation.

## Telescopic, multiplicative, and rational extensions of summations, *with Robert Dawson*

Preprint. arXiv: 2105.04592

A summation is a shift-invariant  $\mathbb{R}$ -module homomorphism from a submodule of  $\mathbb{R}$  to  $\mathbb{R}$  or another ring. Dawson formalized a method for extending a summation to a larger domain by telescoping. In this paper, we revisit telescoping, we study multiplicative closures of summations (such as the usual summation on convergent series) that are not themselves multiplicatively closed, and we study rational extensions as a generalization of telescoping.

## Expository Notes

### Integral-differential calculus

Unpublished. arXiv: 2604.19074

We give an exposition of the Newton-Leibniz calculus. We begin by defining the integral as a limit of Riemann sums, verify the integrals of the standard catalog of functions by direct manipulation, prove the substitution lemmas as theorems about Riemann sums, cross the Fundamental Theorem of Calculus, and harvest the differential calculus on the other side.

### On the symmetry of evidential support

Unpublished. arXiv: 2603.29786

For events  $A$  and  $B$ , we have  $\mathbb{P}(A \mid B) > \mathbb{P}(A \mid \neg B)$  if and only if  $\mathbb{P}(B \mid A) > \mathbb{P}(B \mid \neg A)$  whenever all four quantities are defined.

This note gives seven different proofs of this fact and develops a surrounding web of interpretations.

### Minimalist practical numbers

Unpublished.

A natural number  $n$  is practical if every smaller number can be written as a sum of distinct divisors of  $n$ . We say that a practical number  $n$  is minimalist if this representation is unique. In this note, we prove that a practical number is minimalist if and only if it is a power of 2.

### Fast-growing series are transcendental, *with Robert Dawson*

Unpublished. arXiv: 2102.12995

Let  $R$  be a subring of  $\mathbb{C}[[z]]$ , and let  $X \in \mathbb{C}[[z]]$ . The Newton-Puiseux Theorem implies that if the coefficients of  $X$  grow sufficiently rapidly relative to the coefficients of the series in  $R$ , then  $X$  is transcendental over  $R$ . We provide an alternative proof of this result by establishing a relationship between the coefficients of  $A(X)$  and  $A'(X)$ , where  $A(T)$  is a polynomial over  $\mathbb{C}[[z]]$ .

## Teaching Experience

### Dartmouth College, Hanover, NH (Instructor)

Math 8 (Calculus of One and Several Variables) Fall 2021

Math 1 (Algebra and Calculus) Fall 2020

### Dartmouth College, Hanover, NH (Teacher's Assistant)

Math 100 / Computer Science 49/149 (Random Walk) Spring 2022

Math 100 / Computer Science 49/149 (Decision Theory) Spring 2021

Math 23 (Differential Equations) Winter 2020

Math 3 (Calculus) Fall 2019

Math 22 (Linear Algebra) Spring 2019

Math 22 (Linear Algebra) Fall 2018

### Brigham Young University, Provo, UT (Instructor)

Math 112 (Calculus I) Summer 2017

**Brigham Young University, Provo, UT (Teacher's Assistant)**

Math 112 (Calculus I)

Fall 2017

Math 113 (Calculus II)

Winter 2017

**Brigham Young University, Provo, UT (Grader)**

Math 570 (Matrix Analysis)

Winter 2017

Math 112 (Calculus I)

Fall 2015

## Awards and Achievements

**Certifications**

PSM I Certification

**Awards and Fellowships**

Dartmouth Graduate Fellowship

Fall 2018 - Summer 2023

Gridley Fund for Graduate Mathematics

Fall 2018 - Summer 2019

NSF Graduate Research Fellowship Honorable Mention

April 2018

BYU Academic Scholarship

Spring 2015 - Winter 2016

BYU Math Department Scholarship (Putnam)

Fall 2016

BYU Math Department Scholarship (Putnam)

Fall 2015

BYU Academic Scholarship

Fall 2013 - Winter 2014

**Achievements and Honors**3<sup>rd</sup> Place in the Virginia Tech Regional Math Competition

October 2016

2<sup>nd</sup> Place in the Search for High Rank Elliptic Curve

at 2016 Connecticut Summer School in Number Theory

August 2016

Gold Palm Eagle Scout

December 2012

Black Belt (Taekwondo)

August 2012

**Special Training**

Dale Carnegie Training: Leadership Training for Managers

Summer 2025

Dartmouth Mathematics Teaching Seminar

Spring 2020

BYU Mathematics Teaching Seminar

Fall 2018

## Leadership and Community Service

**Referee***Experimental Mathematics • Integers • Journal of Mathematical Analysis and Applications • Journal of Number Theory • Mathematische Zeitschrift • Theory and Decision***Reviewer***zbMATH Open • CRC Press***K-12 Outreach**

Utah Math Olympiad Committee Member

2015 - Present

Lumiere Mentor

2024

Activity Leader at Dartmouth Science Day (Games &amp; Surreal Numbers)

April 8, 2023

Activity Leader at Lebanon Ward Pi Day

(Trachtenberg Arithmetic &amp; Random Walk)

March 25, 2023

|                                                                               |                   |
|-------------------------------------------------------------------------------|-------------------|
| Session Leader at Dartmouth Sonia Kovalevsky Day<br>(Trachtenberg Arithmetic) | May 21, 2022      |
| Activity Leader at Lebanon Ward Pi Day (Random Walk)                          | April 8, 2022     |
| Exploring Mathematics Camp Leader (Graph Theory)                              | July 27-31, 2020  |
| Exploring Mathematics Camp Leader (Cryptography)                              | July 13-17, 2020  |
| Activity Leader at Lebanon Ward Pi Day (Random Walk)                          | March 7, 2020     |
| Volunteer at Dartmouth Sonia Kovalevsky Day                                   | May 11, 2019      |
| Activity Leader at Dartmouth Science Day (Nim)                                | May 4, 2019       |
| Math Circles Guest Speaker (Tropical Algebra)                                 | February 24, 2018 |
| Math Circles Guest Speaker (Continued Fractions)                              | October 21, 2017  |
| Proofreader for Utah State Math Contest                                       | February 2017     |
| Math Circles Counselor                                                        | 2013 - 2015       |

### **Dartmouth Graduate Student Council**

|                                                  |             |
|--------------------------------------------------|-------------|
| Ad Hoc Healthcare Committee Founder and Co-Chair | 2022 - 2023 |
| ◦ <i>Budget Committee Member</i>                 | Summer 2022 |
| Finance Officer                                  | 2021 - 2022 |
| ◦ <i>Budget Committee Member</i>                 | Summer 2021 |
| Representative for Math Department               | 2020 - 2021 |
| ◦ <i>Service Committee Member</i>                | 2020 - 2021 |
| ◦ <i>Budget Committee Member</i>                 | Summer 2020 |

### **Dartmouth Directed Reading Program**

|                              |             |
|------------------------------|-------------|
| Mentor (Fuzzy Logic)         | Winter 2023 |
| Mentor (Decision Theory)     | Spring 2022 |
| Mentor (Fractional Calculus) | Winter 2022 |

### **Other**

|                                                       |             |
|-------------------------------------------------------|-------------|
| LMFDB Contributor                                     | 2019 - 2023 |
| Dartmouth Algebra and Number Theory Seminar Organizer | 2019 - 2022 |
| CCR-P SCAMP Number Theory Discussion Group Organizer  | Summer 2022 |
| BYU Putnam Team Captain                               | 2014 - 2016 |

## **Thesis Defenses**

|                                                                                                   |                |
|---------------------------------------------------------------------------------------------------|----------------|
| <b>Counting elliptic curves with a cyclic <math>m</math>-isogeny over <math>\mathbb{Q}</math></b> | April 24, 2023 |
| Dartmouth College PhD Thesis Defense                                                              |                |
| <b>The arithmetic of modular grids</b>                                                            | June 22, 2018  |
| BYU Master's Thesis Defense                                                                       |                |

## **Invited Talks**

|                                                                      |                  |
|----------------------------------------------------------------------|------------------|
| <b>Quantum communication &amp; computing</b>                         | January 27, 2025 |
| 2025 ARKA Tech Day                                                   |                  |
| <b>Counting elliptic curves with a cyclic <math>m</math>-isogeny</b> | February 9, 2023 |
| Job Talk at Metron                                                   |                  |
| <b>A family of analogues to the Ramanujan-Robin criterion</b>        | February 2, 2023 |
| Job Talk at Center for Computer Science, Bowie                       |                  |

|                                                                                            |                   |
|--------------------------------------------------------------------------------------------|-------------------|
| <b>Counting elliptic curves with a 7-isogeny</b><br>BYU Colloquium                         | January 19, 2023  |
| <b>Counting elliptic curves with a 7-isogeny</b><br>Simons Collaboration Annual Meeting    | January 11, 2023  |
| <b>Counting 7-isogenies</b><br>2023 Joint Mathematics Meeting                              | January 6, 2023   |
| <b>Counting elliptic curves with a 7-isogeny</b><br>NSA Seminar (Virtual)                  | December 15, 2022 |
| <b>Title redacted</b><br>Next Generation Cryptography at CCRL                              | November 16, 2022 |
| <b>A family of analogues to the Ramanujan-Robin Criterion</b><br>BYU Number Theory Seminar | October 27, 2022  |
| <b>Title redacted</b><br>Briefing to General Paul Nakasone, then-Director of the NSA       | July 26, 2021     |
| <b>Title redacted</b><br>Briefing to Center for Computing Sciences, Bowie                  | June 25, 2021     |

## Other Conference Talks

|                                                                                                                     |                    |
|---------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>A family of analogues to the Robin criterion</b><br>2026 Joint Mathematics Meetings                              | January 6, 2026    |
| <b>Intersecting varieties with transcendental graphs</b><br>Arizona Winter School 2023: Unlikely Intersections      | March 8, 2023      |
| <b>The LCM product and Grönwall's theorem</b><br>2021 Maine-Québec Number Theory Conference (virtual)               | October 3, 2021    |
| <b>Formal summation of divergent series</b><br>2020 Conférence de Théorie des Nombres Québec-Maine (virtual)        | September 26, 2020 |
| <b>The arithmetic of modular grids</b><br>2019 Maine-Québec Number Theory Conference                                | October 5, 2019    |
| <b>The arithmetic of modular grids</b><br>33 <sup>rd</sup> Automorphic Forms Workshop                               | March 8, 2019      |
| <b>The arithmetic of modular grids</b><br>Building Bridges: 4th EU/US Summer School + Workshop on Automorphic Forms | July 17, 2018      |
| <b>Zagier duality in level <math>p</math> modular spaces</b><br>32 <sup>nd</sup> Automorphic Forms Workshop         | March 21, 2018     |
| <b>Zagier duality in level <math>p</math> modular spaces</b><br>2018 BYU Student Research Conference                | March 3, 2018      |
| <b>Zagier duality in level <math>p</math> modular spaces</b><br>Modular Forms are Everywhere Conference             | May 24, 2017       |
| <b>Weakly holomorphic modular forms of level 11</b><br>31 <sup>st</sup> Automorphic Forms Workshop                  | March 7, 2017      |
| <b>Weakly holomorphic modular forms of level 11</b><br>2017 BYU Student Research Conference                         | March 4, 2017      |

|                                                                                                                     |                |
|---------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Congruence relations in modular forms of prime levels greater than 7</b><br>2015 BYU Student Research Conference | March 21, 2015 |
| <b>Residues and independence numbers of graphs</b><br>2014 BYU Student Research Conference                          | March 15, 2014 |

## Selected Seminar Talks

|                                                                                                            |                    |
|------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Attention is all you need</b><br>ARKA Paper Book Club                                                   | March 27, 2026     |
| <b>From last year to next</b><br>ARKA Algorithm Development Tech Titan Presentation                        | February 20, 2026  |
| <b>MLOps</b><br>ARKA Paper Book Club                                                                       | January 13, 2026   |
| <b>Distances on probabilities</b><br>ARKA Artificial Intelligence Tech Titan Presentation                  | December 12, 2025  |
| <b>Are small language models the future of agentic AI?</b><br>ARKA Paper Book Club                         | November 18, 2025  |
| <b>Is chain-of-thought reasoning a mirage?</b><br>ARKA Paper Book Club                                     | October 28, 2025   |
| <b>Algorithms in 2025: a CCCG/WADS retrospective</b><br>ARKA Algorithm Development Tech Titan Presentation | September 12, 2025 |
| <b>Bayesian stochastic blockmodeling</b><br>ARKA Lunch & Learn Series                                      | July 9, 2025       |
| <b>Testing, testing, testing</b><br>ARKA Algorithm Development Tech Titan Presentation                     | June 6, 2025       |
| <b>Procrustes analysis</b><br>ARKA Lunch & Learn Series                                                    | April 9, 2025      |
| <b>Algorithms and computational complexity</b><br>ARKA Algorithm Development Tech Titan Presentation       | February 21, 2025  |
| <b>Robust optimization</b><br>ARKA Algorithm Development Tech Titan Presentation                           | December 5, 2024   |
| <b>Linear programming</b><br>ARKA Algorithm Development Tech Titan Presentation                            | November 21, 2024  |
| <b>The comedy of errors</b><br>ARKA Algorithm Development Tech Titan Presentation                          | October 24, 2024   |
| <b>Applied category theory</b><br>ARKA Algorithm Development Tech Titan Presentation                       | September 12, 2024 |
| <b>Neurofuzziness</b><br>ARKA Artificial Intelligence Tech Titan Presentation                              | June 27, 2024      |
| <b>Fuzzy logic</b><br>ARKA Algorithm Development Tech Titan Presentation                                   | June 6, 2024       |
| <b>Finite automata for the software engineer</b><br>ARKA Algorithm Development Tech Titan Presentation     | April 25, 2024     |
| <b>Quantum computing IV: Shor's algorithm</b><br>ARKA Algorithm Development Tech Titan Presentation        | March 28, 2024     |

|                                                                                                                           |                    |
|---------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Quantum computing III: QML</b><br>ARKA Artificial Intelligence Tech Titan Presentation                                 | March 21, 2024     |
| <b>Spacepower: doctrine for Space Forces</b><br>ARKA SDA & Space Control Tech Titan Presentation                          | February 29, 2024  |
| <b>Quantum computing II: algorithm fundamentals</b><br>ARKA Algorithm Development Tech Titan Presentation                 | February 29, 2024  |
| <b>Quantum computing I: bits and qubits</b><br>ARKA Algorithm Development Tech Titan Presentation                         | February 15, 2024  |
| <b>Neural differential equations</b><br>ARKA Artificial Intelligence Tech Titan Presentation                              | January 25, 2024   |
| <b>Topological data analysis</b><br>ARKA Algorithm Development Tech Titan Presentation                                    | December 7, 2023   |
| <b>Geometric deep learning</b><br>ARKA Artificial Intelligence Tech Titan Presentation                                    | October 19, 2023   |
| <b>Symbolic segmentation II</b><br>ARKA Algorithm Development Tech Titan Presentation                                     | September 14, 2023 |
| <b>Symbolic segmentation I</b><br>ARKA Algorithm Development Tech Titan Presentation                                      | September 14, 2023 |
| <b>The Fox <math>H</math>-function</b><br>Dartmouth Graduate Student Seminar                                              | March 28, 2023     |
| <b>Geometric deep learning</b><br>Dartmouth Graduate Student Seminar                                                      | February 14, 2023  |
| <b>Nearing nearings</b><br>Dartmouth Graduate Student Seminar                                                             | September 14, 2022 |
| <b>Universal algebra and coalgebra</b><br>Dartmouth Graduate Student Seminar                                              | March 30, 2022     |
| <b>Cogalois theory</b><br>Dartmouth Graduate Student Seminar                                                              | February 16, 2022  |
| <b>Cyclic resolution of singularities</b><br>Reading Seminar on Hilbert Modular Surfaces                                  | January 25, 2022   |
| <b>What if the Riemann hypothesis is false?</b><br>Dartmouth Graduate Student Seminar                                     | January 19, 2022   |
| <b>Inverse semigroups: groups without identity</b><br>Dartmouth Graduate Student Seminar                                  | November 9, 2021   |
| <b>Counting 7-isogenies</b><br>Dartmouth Algebra and Number Theory Seminar                                                | November 8, 2021   |
| <b>A sober look at pointless topology</b><br>Dartmouth Graduate Student Seminar                                           | September 21, 2021 |
| <b>Fast-growing series are transcendental</b><br>Dartmouth Graduate Student Seminar (virtual)                             | April 21, 2021     |
| <b>Coalgebras and Hopf algebras</b><br>Reading Seminar on Affine Group Schemes (virtual)                                  | April 15, 2021     |
| <b>Absurd equalities and Runge's method: the degenerate case</b><br>Dartmouth Algebra and Number Theory Seminar (virtual) | March 30, 2021     |

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| <b><math>p</math>-adic Hodge theory</b><br>Reading Seminar on Classical and Quadratic Chabauty (virtual)                    | March 2, 2021      |
| <b>Odd, spoof quasiperfect factorizations</b><br>Dartmouth Algebra and Number Theory Seminar (virtual)                      | February 9, 2021   |
| <b>A primer in social choice theory</b><br>Dartmouth Graduate Student Seminar (virtual)                                     | January 20, 2021   |
| <b>Examples of Kedlaya's algorithm</b><br>Reading Seminar on Classical and Quadratic Chabauty (virtual)                     | November 19, 2020  |
| <b>The LCM product and Grönwall's theorem</b><br>Dartmouth Algebra and Number Theory Seminar (virtual)                      | November 17, 2020  |
| <b>Reactive means and the prisoner's dilemma</b><br>Dartmouth Graduate Student Seminar (virtual)                            | October 7, 2020    |
| <b>Formal summation of divergent series: an algebraic approach</b><br>Dartmouth Algebra and Number Theory Seminar (virtual) | April 28, 2020     |
| <b>On the infinitude of the natural numbers</b><br>Dartmouth Graduate Student Seminar                                       | February 12, 2020  |
| <b>Variations of Hodge structures</b><br>Reading Seminar on Shimura Varieties                                               | November 26, 2019  |
| <b>Hodge structures</b><br>Reading Seminar on Shimura Varieties                                                             | November 14, 2019  |
| <b>Savage's expected utility and making good decisions</b><br>Dartmouth Graduate Student Seminar                            | November 13, 2019  |
| <b>Geometric and generalized calculus</b><br>Dartmouth Graduate Student Seminar                                             | October 8, 2019    |
| <b>Summing divergent series</b><br>Dartmouth Graduate Student Seminar                                                       | July 31, 2019      |
| <b>Real analysis: a nonstandard approach</b><br>Dartmouth Graduate Student Seminar                                          | April 17, 2019     |
| <b>Why save the universe? Set theory with a universal set</b><br>Dartmouth Graduate Student Seminar                         | January 16, 2019   |
| <b>The arithmetic of modular grids</b><br>Dartmouth Graduate Student Seminar                                                | September 26, 2018 |
| <b>Dioids and idempotent geometry II</b><br>BYU Algebra Seminar                                                             | March 13, 2018     |
| <b>Dioids and idempotent geometry I</b><br>BYU Algebra Seminar                                                              | February 27, 2018  |
| <b>Zagier duality in level <math>p</math> modular spaces</b><br>BYU Number Theory Seminar                                   | February 22, 2018  |
| <b>Average values of arithmetic functions</b><br>BYU Number Theory Seminar                                                  | October 12, 2017   |
| <b>Weakly holomorphic modular forms of level 11</b><br>BYU Number Theory Seminar                                            | February 9, 2017   |

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| <b>Generating functions for canonical bases of certain level 11 weakly holomorphic modular forms</b><br>BYU Number Theory Seminar | January 26, 2017  |
| <b>High rank elliptic curves with prescribed torsion</b><br>BYU Number Theory Seminar                                             | December 1, 2016  |
| <b>Heuristics for elliptic curves of high rank</b><br>BYU Number Theory Seminar                                                   | October 20, 2016  |
| <b>A lemma regarding the Feit-Thompson conjecture</b><br>BYU Number Theory Seminar                                                | November 12, 2015 |
| <b>Algebraic number theory and the Feit-Thompson conjecture</b><br>BYU Number Theory Seminar                                      | October 1, 2015   |
| <b>Congruence relations in modular forms of prime levels greater than 7</b><br>BYU Number Theory Seminar                          | March 19, 2015    |
| <b>Hensel's lemma</b><br>BYU Number Theory Seminar                                                                                | February 12, 2015 |
| <b>Newton polynomials</b><br>BYU Number Theory Seminar                                                                            | March 27, 2014    |

## Conferences and Workshops Attended

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| <i>2026 Joint Mathematics Meetings</i><br>Walter E. Washington Convention Center, Washington DC                                                                                       | January 4-7, 2026             |
| <i>CCCG/WADS 2025</i><br>York University, Toronto                                                                                                                                     | August 11 - 15, 2025          |
| <i>Qubits 2025: Quantum Realized</i><br>Virtual (Scottsdale, Arizona)                                                                                                                 | March 31 - April 1, 2025      |
| <i>2025 Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation Annual Meeting</i><br>Virtual (Simons Foundation's Gerald D. Fischbach Auditorium, New York City) | January 15-16, 2025           |
| <i>2024 Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation Annual Meeting</i><br>Virtual (Simons Foundation's Gerald D. Fischbach Auditorium, New York City) | January 11-12, 2024           |
| <i>Big Data Conference 2023</i><br>Virtual (Harvard University, Cambridge)                                                                                                            | August 31 - September 1, 2023 |
| <i>Hilbert Modular Forms Infrastructure Week 4</i><br>Dartmouth College, Hanover                                                                                                      | May 8-12, 2023                |
| <i>Arizona Winter School 2023: Unlikely Intersections</i><br>University of Arizona, Tucson                                                                                            | March 4-8, 2023               |
| <i>2023 Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation Annual Meeting</i><br>Simons Foundation's Gerald D. Fischbach Auditorium, New York City           | January 11-12, 2023           |
| <i>2023 Joint Mathematics Meetings</i><br>Hynes Convention Center, Boston                                                                                                             | January 4-7, 2023             |

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| <i>Hilbert Modular Forms Infrastructure Week 3</i><br>Dartmouth College, Hanover                                                                                                        | December 12-16, 2022   |
| <i>Next-Generation Cryptography</i><br>Center for Communications Research, La Jolla                                                                                                     | November 14-18, 2022   |
| <i>Modular Curves Workshop 2</i><br>MIT, Boston                                                                                                                                         | November 5-9, 2022     |
| <i>Big Data Conference 2022</i><br>Virtual (Harvard University, Cambridge)                                                                                                              | August 26, 2022        |
| <i>Graph Fest 2022</i><br>NSA Internal Conference                                                                                                                                       | June 29-30, 2022       |
| <i>Explicit Methods for Modularity</i><br>Virtual (Simons Collaboration)                                                                                                                | April 11-15, 2022      |
| <i>Arizona Winter School 2022: Automorphic Forms Beyond <math>GL_2</math></i><br>Virtual (University of Arizona, Tucson)                                                                | March 5-9, 2022        |
| <i>Hilbert Modular Forms Infrastructure Week 2</i><br>Dartmouth College, Hanover                                                                                                        | February 21-25, 2022   |
| <i>2021 Maine-Québec Number Theory Conference</i><br>Virtual (University of Maine, Orono)                                                                                               | October 2-3, 2021      |
| <i>MathFest 2021</i><br>NSA Internal Conference                                                                                                                                         | June 4, 2021           |
| <i>Spring 2021 Algebraic Geometry Northeastern Series</i><br>Virtual (Brown University, Providence)                                                                                     | May 4-5, 2021          |
| <i>21<sup>st</sup> Algebra, Geometry and Combinatorics Day</i><br>Virtual (University of Notre Dame, Notre Dame)                                                                        | April 10, 2021         |
| <i>2021 Joint Mathematics Meetings</i><br>Virtual (American Math Society, Mathematical Association of America)                                                                          | January 6-9, 2021      |
| <i>Fall 2020 Algebraic Geometry Northeastern Series</i><br>Virtual (Stony Brook University, Stony Brook)                                                                                | October 23-25, 2020    |
| <i>2020 Conférence de Théorie des Nombres Québec-Maine</i><br>Virtual (Université Laval, Québec)                                                                                        | September 26-27, 2020  |
| <i>14<sup>th</sup> Algorithmic Number Theory Symposium</i><br>Virtual (University of Auckland, Auckland)                                                                                | June 29 - July 4, 2020 |
| <i>UNCG Summer School in Computational Number Theory and Algebra:<br/>Ergodic Theory with Applications to Continued Fractions</i><br>Virtual (University of North Carolina, Greensboro) | May 18-22, 2020        |
| <i>2019 Maine-Québec Number Theory Conference</i><br>University of Maine, Orono                                                                                                         | October 5-6, 2019      |
| <i>Fall 2019 Algebraic Geometry Northeastern Series</i><br>Boston College, Boston                                                                                                       | September 20-22, 2019  |
| <i>33<sup>rd</sup> Automorphic Forms Workshop</i><br>Duquesne University, Pittsburgh                                                                                                    | March 6-10, 2019       |

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| <i>Building Bridges: 4<sup>th</sup> EU/US Summer School +<br/>Workshop on Automorphic Forms and Related Topics</i><br>Alfréd Rényi Institute of Mathematics, Budapest | July 9-20, 2018   |
| <i>32<sup>nd</sup> Automorphic Forms Workshop</i><br>Tufts University, Medford                                                                                        | March 19-22, 2018 |
| <i>Modular Forms are Everywhere</i><br>Max Planck Institute for Mathematics, Bonn                                                                                     | May 15-26, 2017   |
| <i>31<sup>st</sup> Automorphic Forms Workshop</i><br>East Tennessee State University, Johnson City                                                                    | March 6-9, 2017   |
| <i>2017 BYU Student Research Conference</i><br>Brigham Young University, Provo                                                                                        | March 4, 2017     |
| <i>2016 Connecticut Summer School in Number Theory</i><br>University of Connecticut, Storrs                                                                           | August 8-14, 2016 |
| <i>2015 BYU Student Research Conference</i><br>Brigham Young University, Provo                                                                                        | March 21, 2015    |
| <i>2014 BYU Student Research Conference</i><br>Brigham Young University, Provo                                                                                        | March 15, 2014    |